

	Assignment No. 02 SEMESTER Fall 2013 CS301- Data Structures	Total Marks: 20 Due Date: 28/12/2013
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Instructions

Please read the following instructions carefully before solving & submitting assignment:

It should be clear that your assignment will not get any credit (zero marks) if:

- **The assignment is submitted after due date.**
- **The submitted code does NOT compile.**
- **The submitted assignment file is other than .CPP format.**
- **The submitted assignment file does not open or corrupted.**
- **The assignment is copied (from other student or ditto copy from handouts or internet).**

Uploading instructions

For clarity and simplicity, You are required to Upload/Submit only one **.CPP** file.

- **Don't wait for grace day. Grace Day is given only if there is problem with LMS on due date. Submit your solution within due date.**
- **Note that no assignment will be accepted through email if there is any problem on grace day.**

Note: Use only dev-C++ IDE.

Objective

The objective of this assignment is

- To make you familiar with queue and linked list data structures and programming techniques to implement queue as linked list.

For any query about the assignment, contact at cs301@vu.edu.pk

GOOD LUCK

Marks: 20

Suppose there is a large busy cosmetic store. In store, for billing, there are four counters with four tellers (who deal payment etc). A customer comes to the store and collects the things he/she wants to purchase and go to one of the counters to pay for purchase. If there are already people in front of counters, customer has to stand in the queue. A customer generally selects the shortest queue (with respect to the number of people in queue). The selection can become wrong if one of the people in the queue has a huge buying list. But if we know the estimated time of each customer in the queue, selection can become efficient. The system can become efficient, if we tell a new person which one is the shortest queue with respect to time taken by the people waiting in the queue.

Considering the above scenario, write a C++ program which asks a user to enter the estimated time to process the bill. The program will tell the user, in which queue he/she can stand with respect to the total time of all the

persons already in the queues. At the end of the day the system will display the total number of customers deal by every Teller and which Teller has deal maximum number of customers.

Solution Guidelines:

- 1) To pay the bills, Customer must have to stand in the queue. The length of the queue can vary so you have to implement queue as linked list.
- 2) **Create a class named Customer (Node class);** this class should create a node (customer) in the queue. This class should define two private data members, time and Next pointer. A constructor, getter and setter functions for each data member should also be defined in this class.
- 3) **Create another class named Teller (List class);** to create a queue for each teller, this class should define three variables, Front pointer, Rear pointer and counter. The class should also define the following functions:

Constructor(): Default constructor of the class.

Destructor(): Destructor to destroy the memory gained by the use of new operator.

AddCustomer(): This method should enter the new customer at the rear.

TotalCustomer(): This method should return the total number of customers in a queue.

TotalTime(): This method should return the total time of customers standing in said queue.

- 4) In main() method of the program, create four objects of the class Teller which will act like four counters at store. These objects should be able to perform all the required functionalities mentioned in the question statement.
- 5) You can add more variables and functions to obtain the given functionalities according to your need.

Sample Output

```
Enter duration of your transaction5
You can stand in Queue 1

Do you Want to Buy Another Ticket<Y/N>y

Enter duration of your transaction2
You can stand in Queue 2

Do you Want to Buy Another Ticket<Y/N>y

Enter duration of your transaction4
You can stand in Queue 3

Do you Want to Buy Another Ticket<Y/N>y

Enter duration of your transaction3
You can stand in Queue 4

Do you Want to Buy Another Ticket<Y/N>y

Enter duration of your transaction1
You can stand in Queue 2

Do you Want to Buy Another Ticket<Y/N>n

Number of Customers Deal By Every Teller
T1=      1
T2=      2
T3=      1
T4=      1

TELLER 2 Deal Maximum Customers of the Day

Press any key to continue . . .
```

Lectures Covered: This assignment covers Lecture # 1-10

Deadline: Your assignment must be uploaded / submitted on / before, 28 November, 2013.